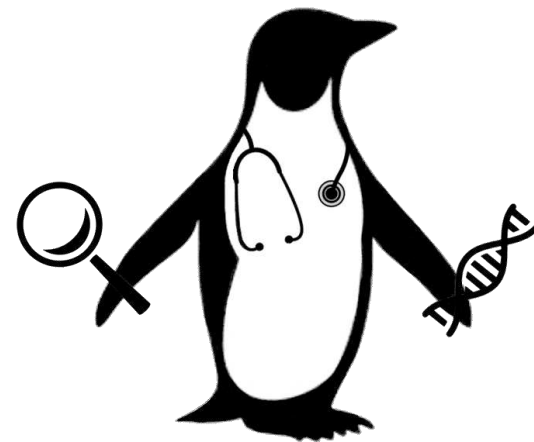




PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice

& other RDF-related fun

Elias Crum

SwissProt Presentation
March 25th, 2025

Personal Genome Data Usage?

My Journey:



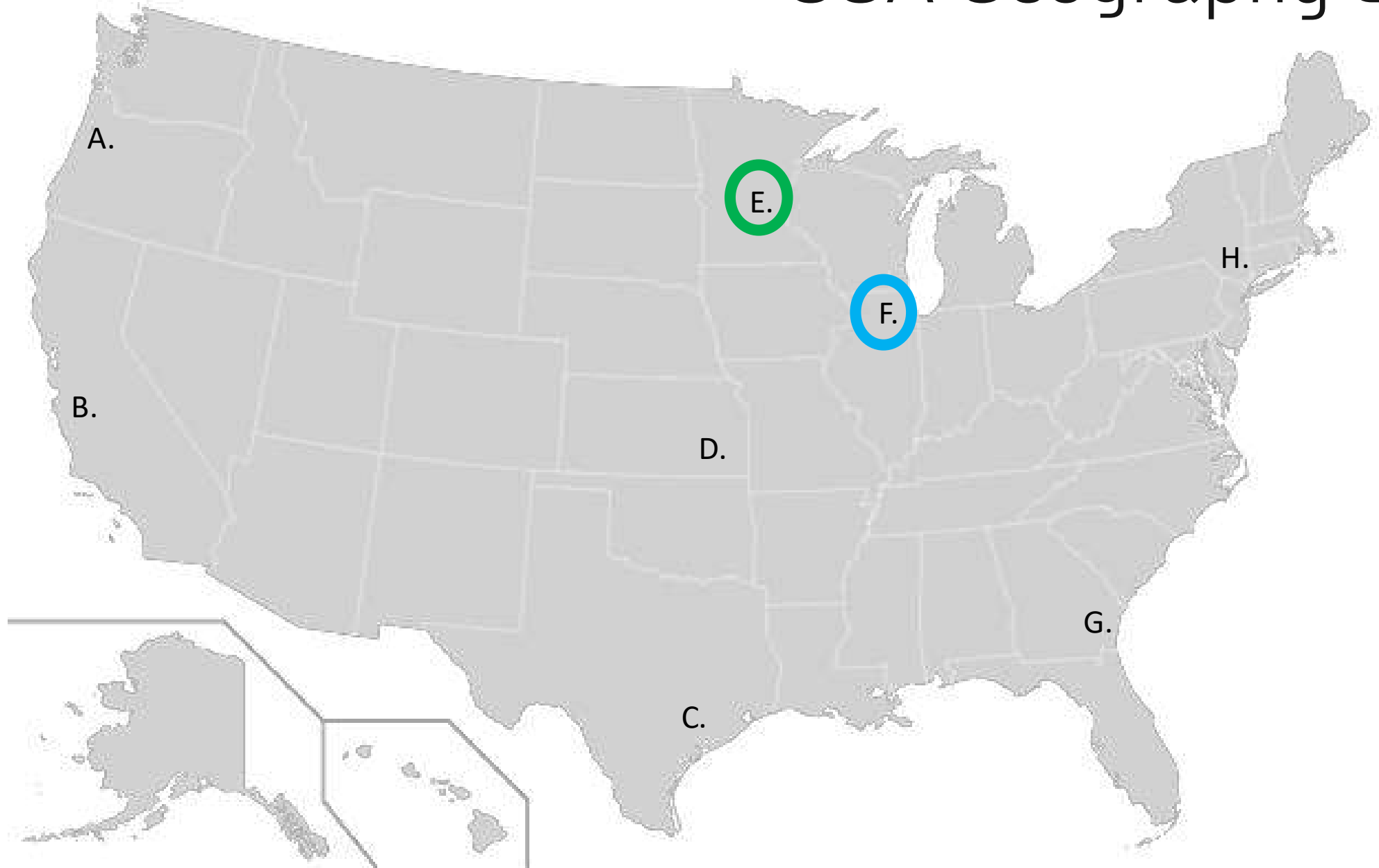
BSc + MSc - Bioinformatics



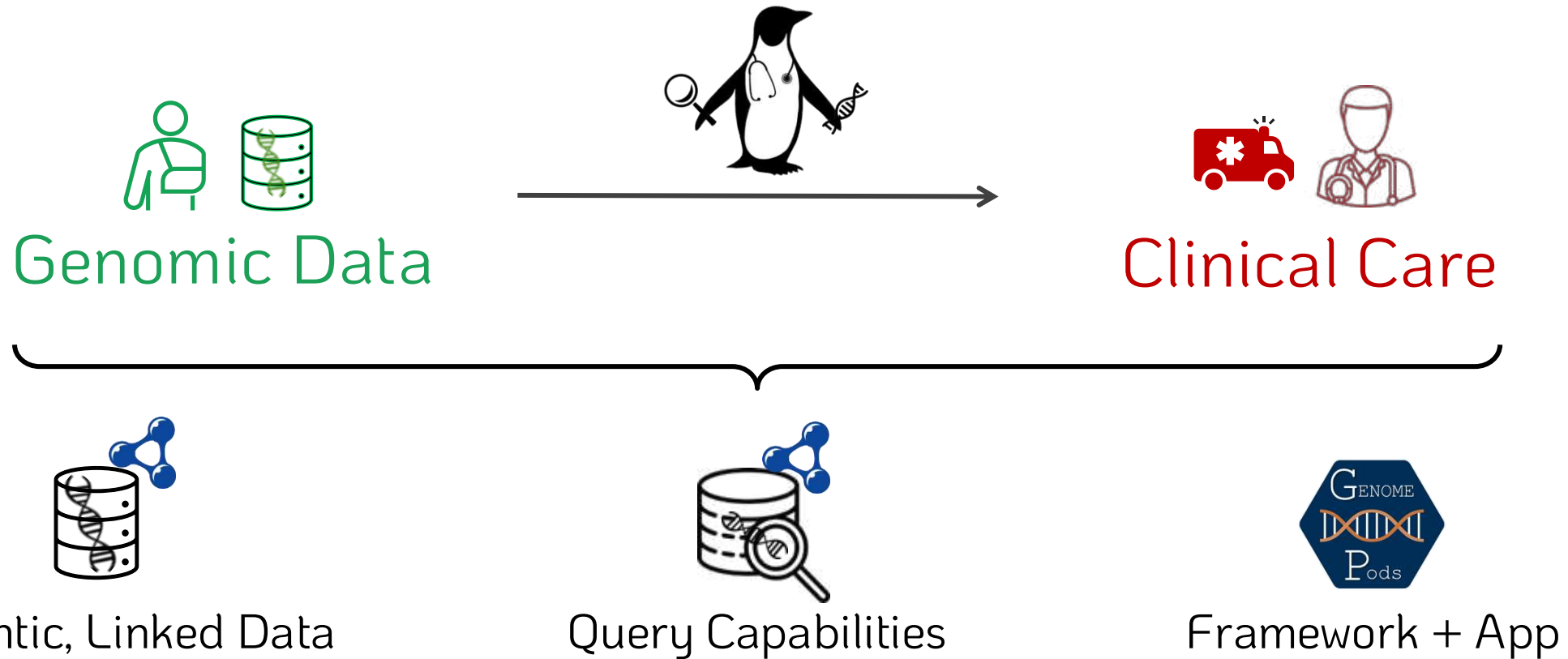
Prokaryotic Genomics

How can we increase **genomics** use in **medicine**?

USA Geography Lesson



PENGQUIN



Promoters: Dr. Gökhan Ertaylan¹, Dr. Bart Buelens¹, Dr. Ruben Taelman², Prof. Dr. Ruben Verborgh²

¹ = VITO NV

² = Ghent University

PENGQUIN Goals

Improve genomic data **USAGE** and **SCALABILITY**



Improve genomic data store accessibility (while maintaining privacy)



Represent genomic data semantically as Linked Data



Query genomic data and linkages



Connect to existing initiatives / applications

Proof-of-concept



Develop a framework and web application

Talk Outline



Decentralized Data Storage (*CHIST-ERA TRIPLE Project*)



Introduction to the Project



Introduction to Solid Pods



The Solid Cockpit web application (and maybe a use-case example if we have time)



Federated Querying (*Real World Federated Query Survey*)



Motivation + Challenges



Progress and preliminary findings



Future aims

CHIST-ERA TRIPLE Project



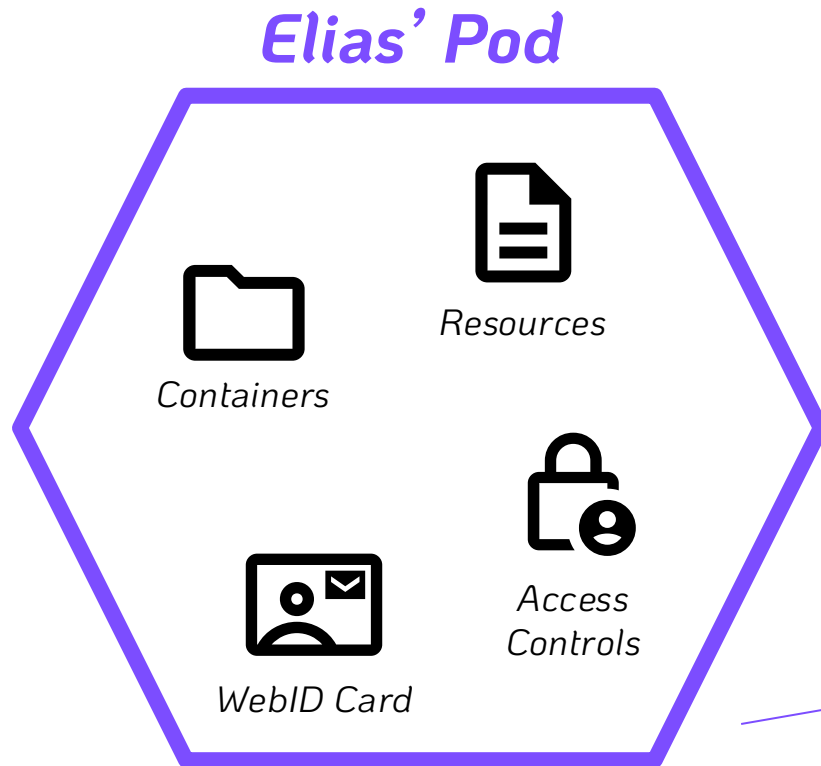
**Standardized documentation
and resource statistics**

What is a Solid Pod?

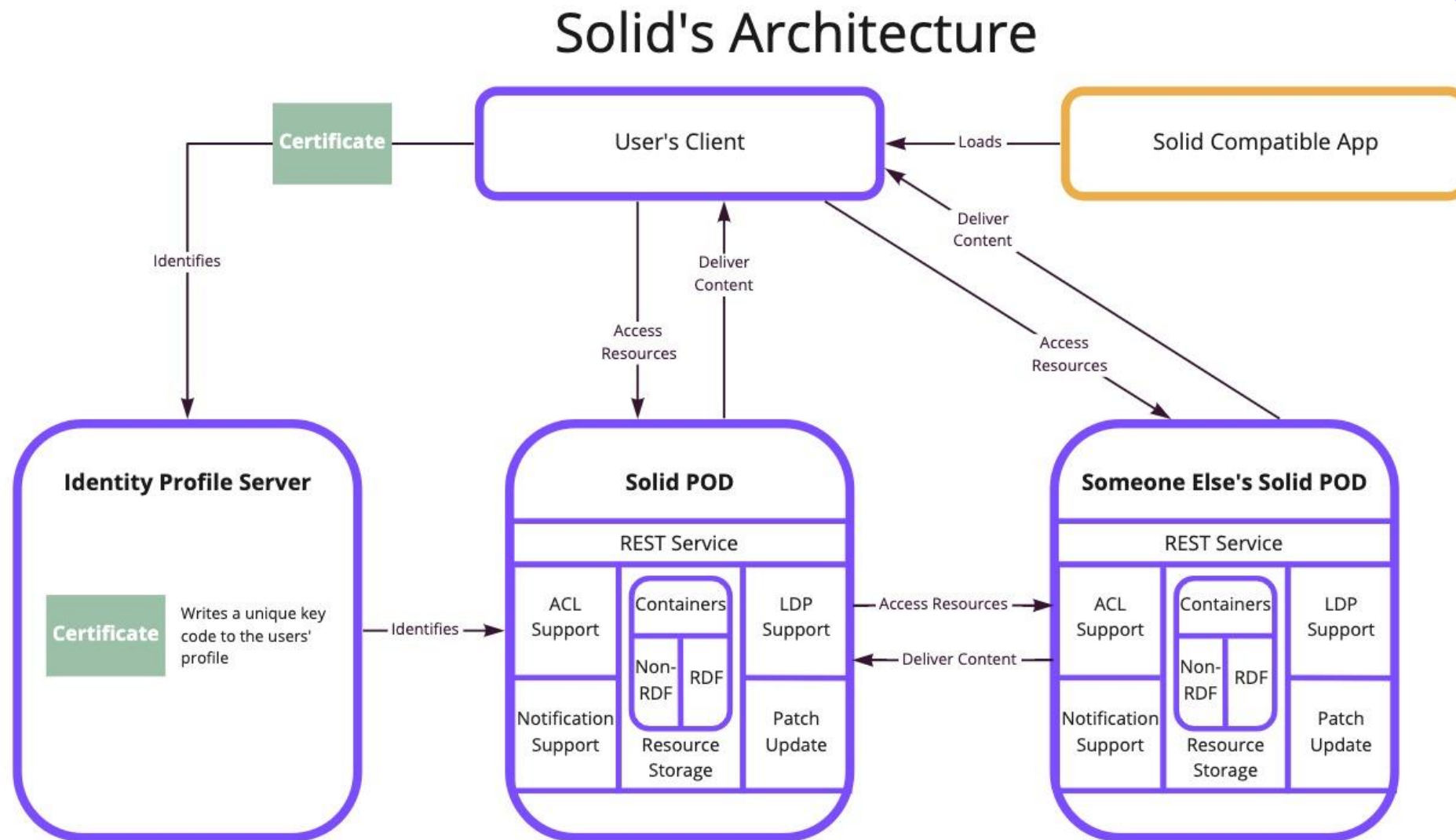


Secure, Decentralized Data Storage Specification

- ↳ *Pod exists on a Server (accessible to the Web)*
- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*



What is a Solid Pod? - Detailed



miro

Solid Pod vs SPARQL Endpoint?



Solid Pod

- + *decentralized storage strategy*
- + *can store both file data and RDF data*
- + *granular privacy policy capabilities*
- *native querying capabilities limited*
- *not designed for large-scale data*
- *no computation attached to data store*



SPARQL Endpoint

- *centralized storage strategy*
- *triple store (only RDF)*
- *limited privacy capabilities*
- + *designed for high-performance querying*
- + *excels with large datasets*
- + *computation directly attached to data store*

How to query a Solid Pod?

A Query Engine



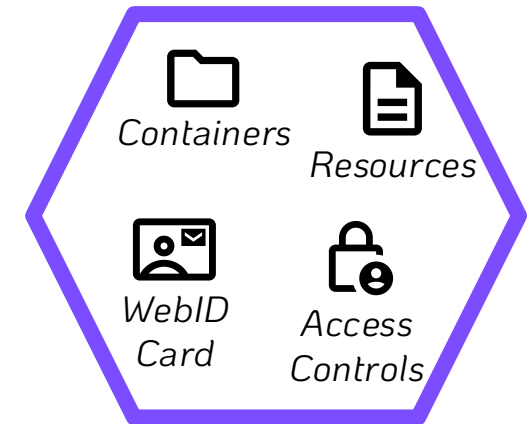
- Query computation client-side
- Allows for SPARQL query execution over Solid Pod API
- Linked Data traversal + heterogeneous result joining capabilities

SPARQL Query

```
1 SELECT DISTINCT ?p {  
2   ?s ?p ?o  
3 } LIMIT 10
```



Elias' Pod



Results Binding Stream
SPARQL+JSON



Solid Pod Querying – Practical Example Concept

1. A research partner has a [proprietary] list of CAS numbers from biological sample analysis...

CAS Common Chemistry

Return to Results

Caffeine

CAS Registry Number[®]
58-08-2

CN1C=NC2=C1C(=O)N(C)C(=O)N2C

CAS Name
Caffeine

Molecular Formula
C₈H₁₀N₄O₂

Molecular Mass
194.19

Discover more in Scifinder[®]

Cite this Page
Caffeine. CAS Common Chemistry. CAS, a division of the American Chemical Society, n.d. https://commonchemistry.cas.org/detail/cas_rn=58-08-2 (retrieved 2022-01-24) (CAS RN: 58-08-2). Licensed under the Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0).

Compound Properties

Melting Point (1)
238 °C

Density (2)
1.23 g/cm³ @ Temp: 18 °C

Source(s)
(1) International Chemical Safety Cards data were obtained from the National Institute for Occupational Safety and Health (US)
(2) Hazardous Substances Data Bank data were obtained from the National Library of Medicine (US)

Other Names and Identifiers

InChI
InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)(7(13)12(3)(8)14)11(8)2/h4,11-13H3

InChIKey
InChIKey=RYYVLZLVUJYJGH-UHFFFAOYSA-N

SMILES
O=C1C2=C(N(C)C(=O)N1C)N=CN2C

Canonical SMILES
O=C1C2=C(N=CN2C)N(C)=O/N1C/C

Other Names for this Substance

- 1H-Purine-2,6-dione, 3,7-dihydro-1,3,7-trimethyl-
- Caffeine
- 3,7-Dihydro-1,3,7-trimethyl-1H-purine-2,6-dione
- Guaranine
- Methyltheobromine

Deleted or Replaced CAS Registry Numbers
95789-13-2, 71701-02-5

2. We want to find other chemicals with similar molecular structures for a toxicity study

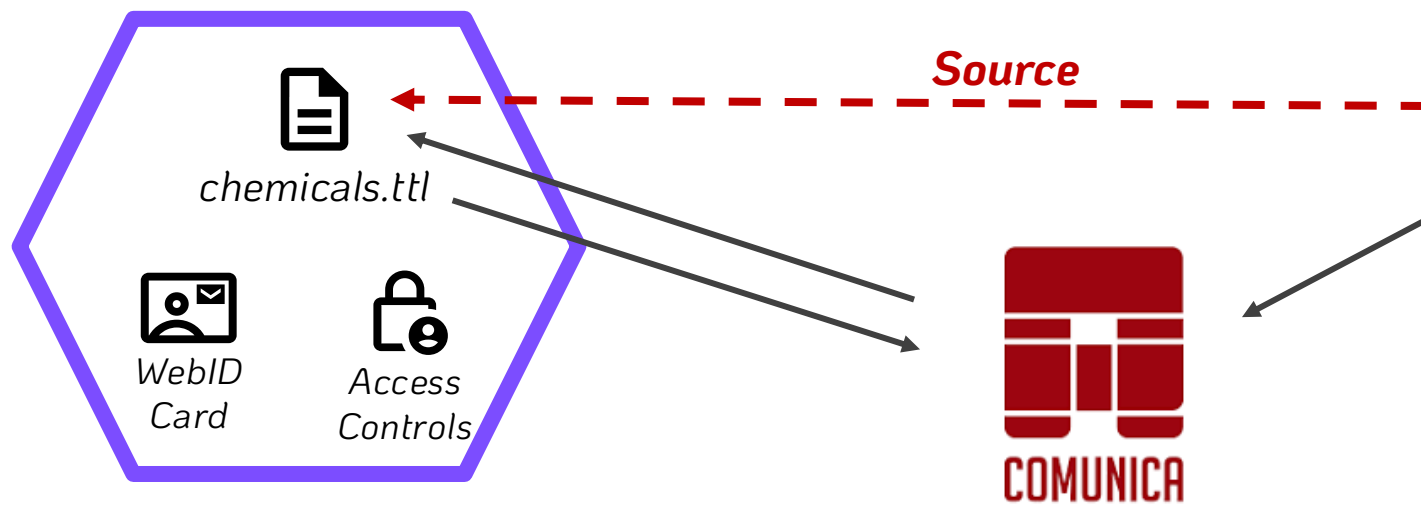
3. How could we design and execute a SPARQL query to achieve this goal?

Solid Pod Querying – Practical Example

SPARQL Query

```
1 prefix sio: <http://semanticscience.org/resource/>
2 PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
3 PREFIX sio: <http://semanticscience.org/resource/>
4 PREFIX vocab: <http://rdf.ncbi.nlm.nih.gov/pubchem/vocabulary#>
5 PREFIX sachem: <http://bioinfo.uochb.cas.cz/rdf/v1.0/sachem#>
6 PREFIX endpoint: <https://idsm.elixir-czech.cz/sparql/endpoint/>
7
8 SELECT ?CAS ?COMPOUND ?SIMILAR_COMPOUND ?SCORE WHERE {
9   ?s sio:SIO_000300 ?CAS .
10  |
11  SERVICE endpoint:idsm {
12    ?SYNONYM a sio:CHEMINF_000446. # CAS registry number
13    ?SYNONYM sio:SIO_000300 ?CAS.
14    ?SYNONYM sio:SIO_000011 ?COMPOUND.
15    ?COMPOUND a vocab:Compound.
16    ?ATTRIBUTE a sio:SIO_011120. # molecular structure file
17    ?ATTRIBUTE sio:SIO_000011 ?COMPOUND.
18    ?ATTRIBUTE sio:SIO_000300 ?MOLFILE.
19  }
20  SERVICE endpoint:chebi {
21    [ sachem:compound ?SIMILAR_COMPOUND;
22      sachem:score ?SCORE ] sachem:similaritySearch [
23        sachem:query ?MOLFILE;
24        sachem:cutoff "0.7"^^xsd:double;
25        sachem:similarityRadius '3'^^xsd:integer;
26        sachem:aromaticityMode sachem:aromaticityDetectIfMissing;
27        sachem:tautomerMode sachem:inchiTautomers
28      ].
29    }
30  }
31  UNION
32  {
33    SERVICE endpoint:chebi {
34      ?SIMILAR_COMPOUND sachem:substructureSearch [ sachem:query '[As]' ].
35    }
36  }
37 }
38 }
39 ORDER BY DESC(?SCORE)
```

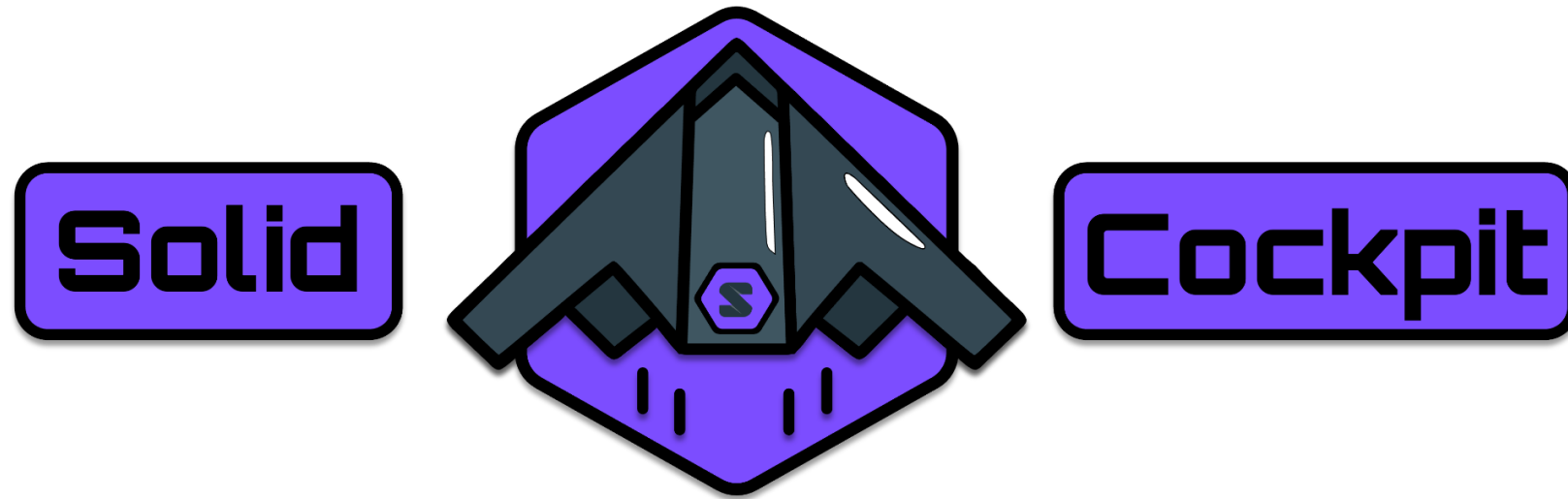
Research Partner's Pod



Results

	CAS	COMPOUND	SIMILAR_COMPOUND	SCORE
0	92-67-1	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_1784	1.0
1	34256-82-1	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_2394	1.0
2	485-31-4	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_83371	1.0
3	485-31-4	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_83367	1.0
4	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_4034	1.0
5	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39313	1.0
6	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39309	1.0
7	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39310	1.0
8	1820573-27-0	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_39312	1.0
9	121-14-2	http://rdf.ncbi.nlm.nih.gov/pubchem/compound/C...	http://purl.obolibrary.org/obo/CHEBI_920	1.0

My Contributions



knowledgeonwebscale.github.io/solid-cockpit/

My Contributions (specifics)

[knowledgeonwebscale
.github.io/solid-cockpit/](https://knowledgeonwebscale.github.io/solid-cockpit/)



A Web Application to:

 *Upload Data to a Solid Pod*

 *Browse contents of a Solid Pod*

 *Edit Privacy of (and share) data in a Solid Pod*

 *Query Data in Solid Pods + SPARQL Endpoints*

↳ *Implement a query cache for storing queries and results*

Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

① Performance is currently ~variable~

↳ We want to detail *WHY* performance varies so much && suggestions for improvement

② Assess federation algorithms on [REAL endpoints](#)

↳ [FedX](#) – uses ASK queries to inform source-selection

↳ [SPLendid](#) – uses VOID descriptions to inform source-selection

Real World Federated Query Survey - Design

SPARQL Endpoints:



“Real World” Federated Queries:



<https://sib-swiss.github.io/sparql-examples/>

SPARQL Query Engine:



<https://comunica.dev/>

Real World Federated Query Survey – [Preliminary] Findings

Preliminary Results (endpoints and queries):

Federated querying is not "easy"

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from <https://semantic.eea.europa.eu/sparql> (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from <https://sparql.orthodb.org/sparql> (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from <https://query.wikidata.org/sparql> (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – [Preliminary] Findings

Preliminary Results (algorithms):

Federated querying is not "easy"

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve <https://idsm.elixir-czech.cz/sparql/endpoint/chebi> (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Real World Federated Query Survey – Future Work

Future Aims:

Execute 1 month worth of queries

↳ 4 experimental runs per week

Vary time + part of week experiment is performed

Publication goals:

- In-Use Paper
- Journal Paper Extension (benchmark creation)

Algorithmic variation

- VOID vs ASK queries
- Various join algorithm combinations
- Rate-limiting
- Service-clause removal

Metrics Assessed

- Total execution time
- Query planning time
- Result set consistency
- # SPARQL requests
- # HTTP requests
- Individual query arrival times

Thank you!



More PhD specifics



LinkedIn

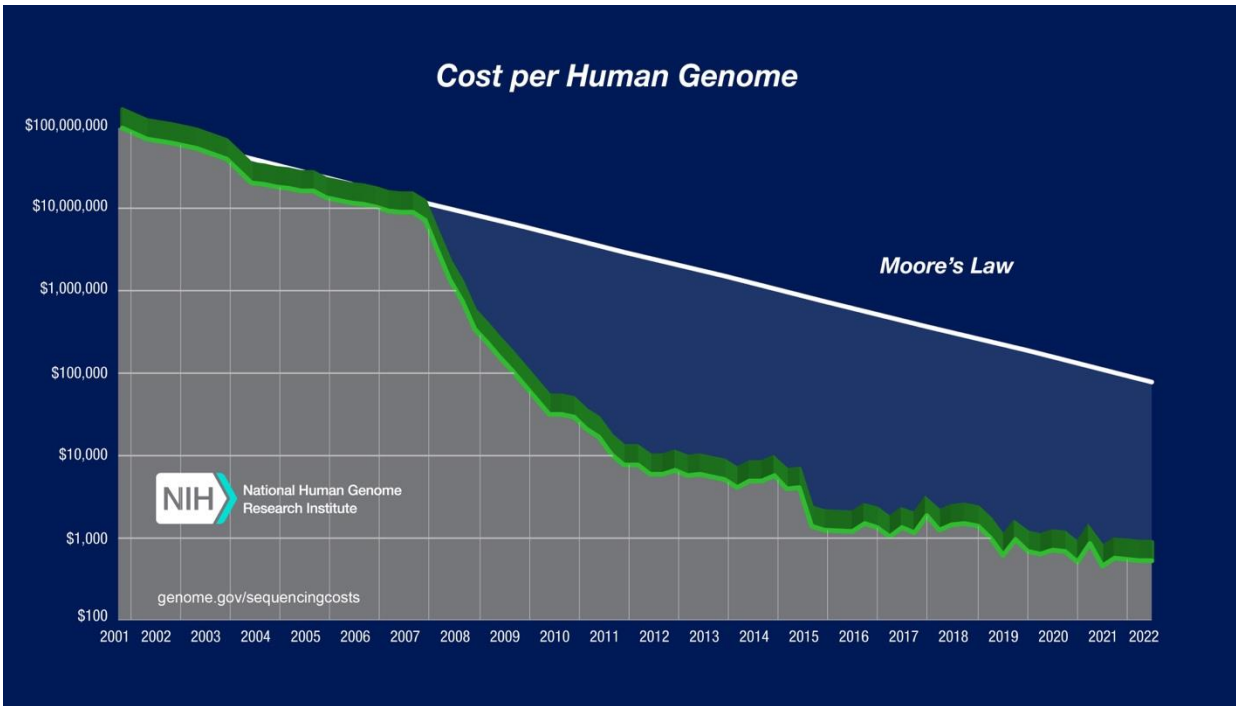


GitHub

Supplemental Slides

Genome Sequencing in Health Care

Sequence Generation Costs



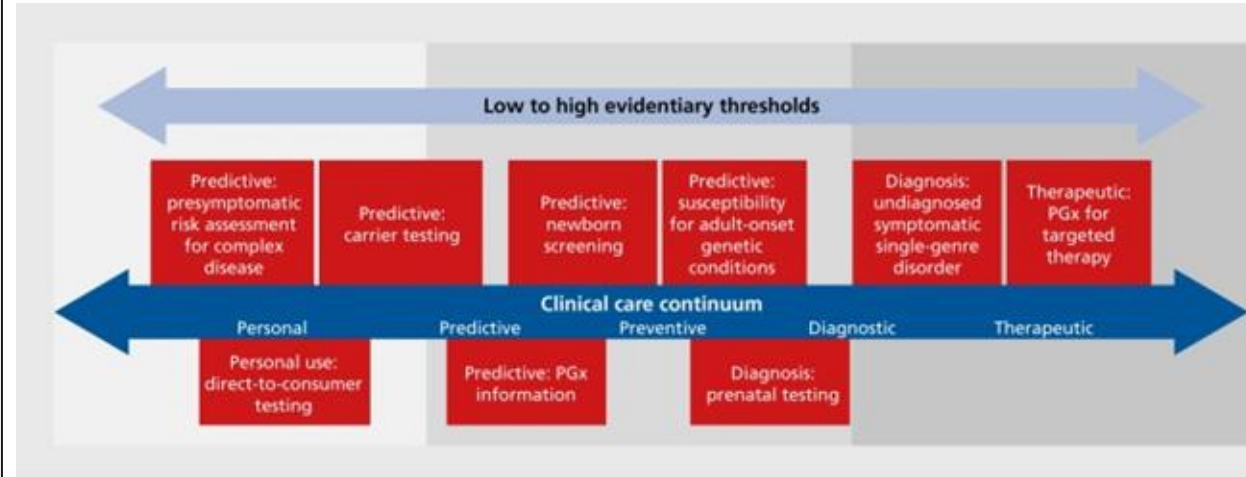
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to \$100 per genome⁴

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Scaling Genome Data Usage

Research genomic data sharing initiatives:



State-of-the-art for **clinical** genomic data sharing?

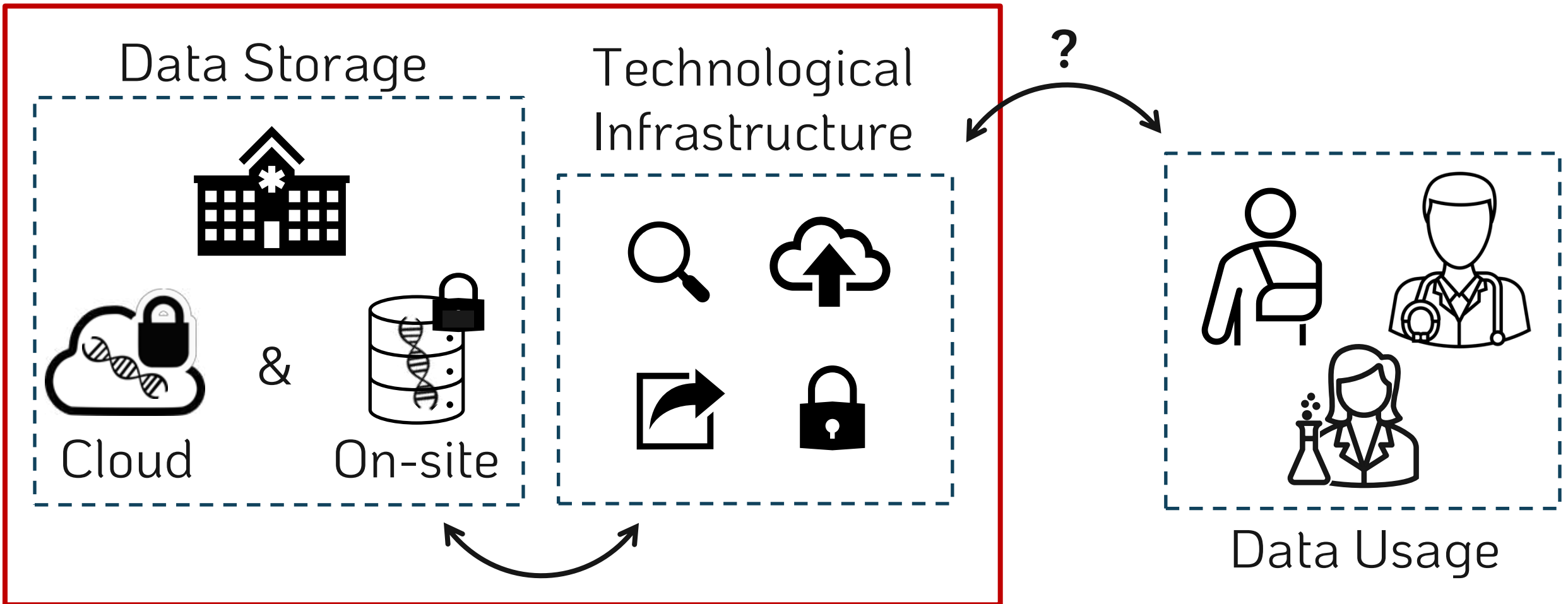
GDPR (EU) / HIPPA (US)
EU Health Data Spaces



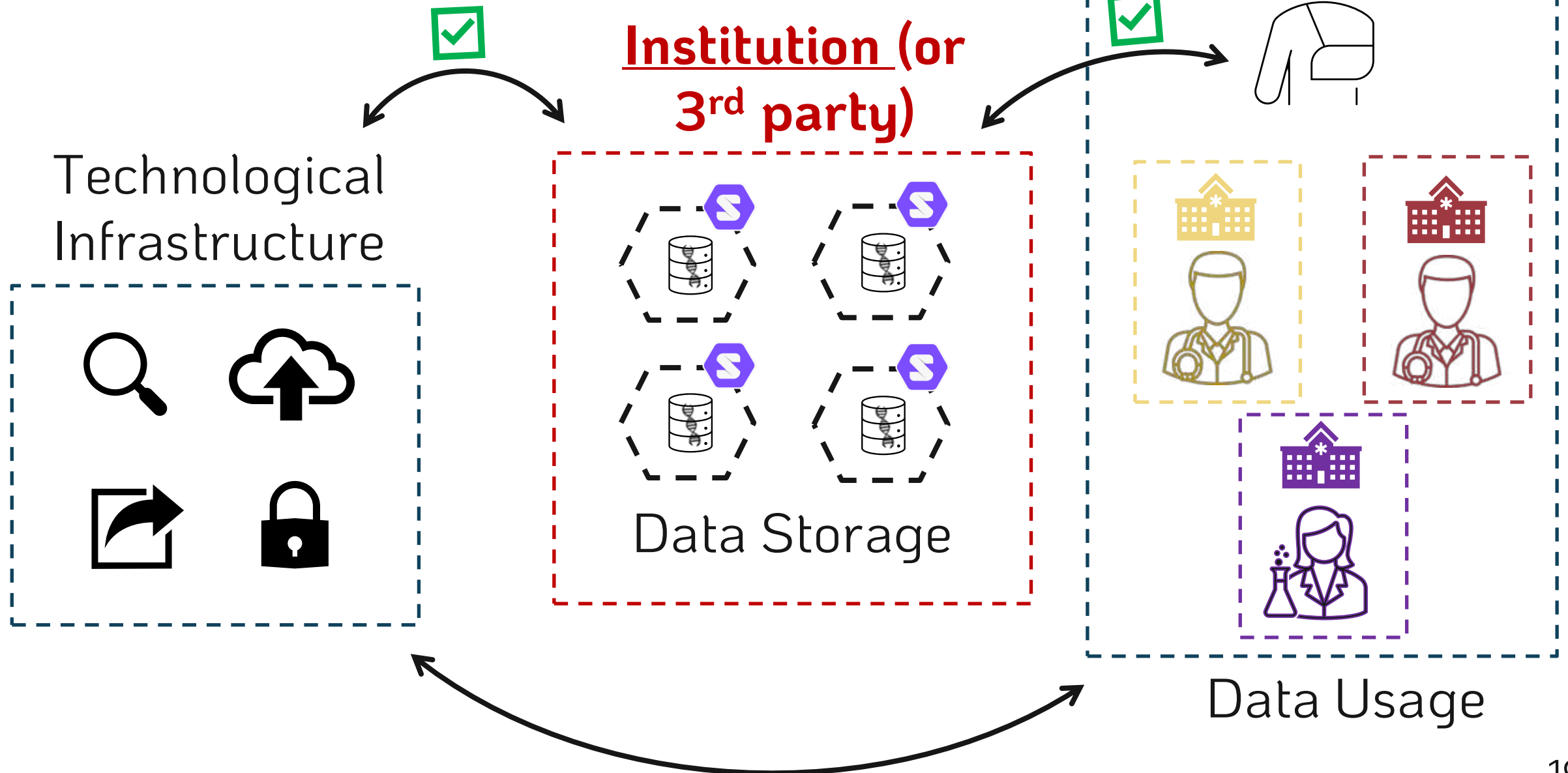
Not much...

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution





- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

Project Goals



Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

PhD Goals

- ✓ Finished
- ⌚ Soon
- 🕒 Future

Basic Functionalities:

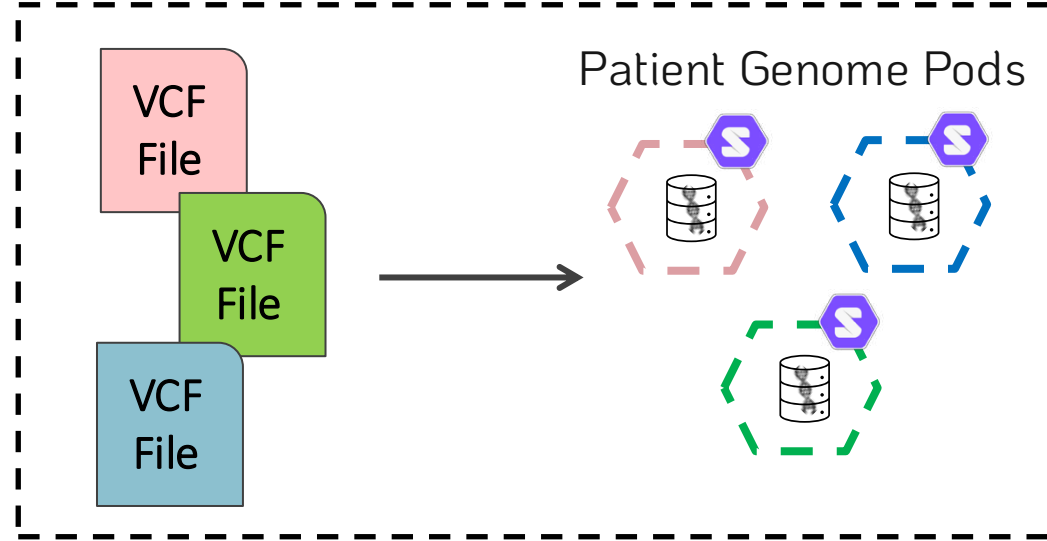
-  Genomic data storage in Solid pods ✓
-  Data privacy management ⌚
-  Data Sharing via the web ✓

Experimental research:

-  Represent genomic data as RDF? ⌚
-  Linking genomic and health data? 🕒
-  Single and Many pod Querying? 🕒

Workflow Overview

⚙️ Data Storage



Data Storage



Solid can be implemented for genomic data storage.



Solid pod instances → Community Solid Server¹

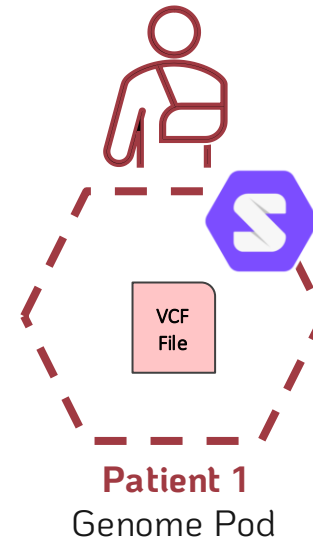


VCF Genomic Data → Illumina Platinum Genomes²

Patient 1
Genome Data

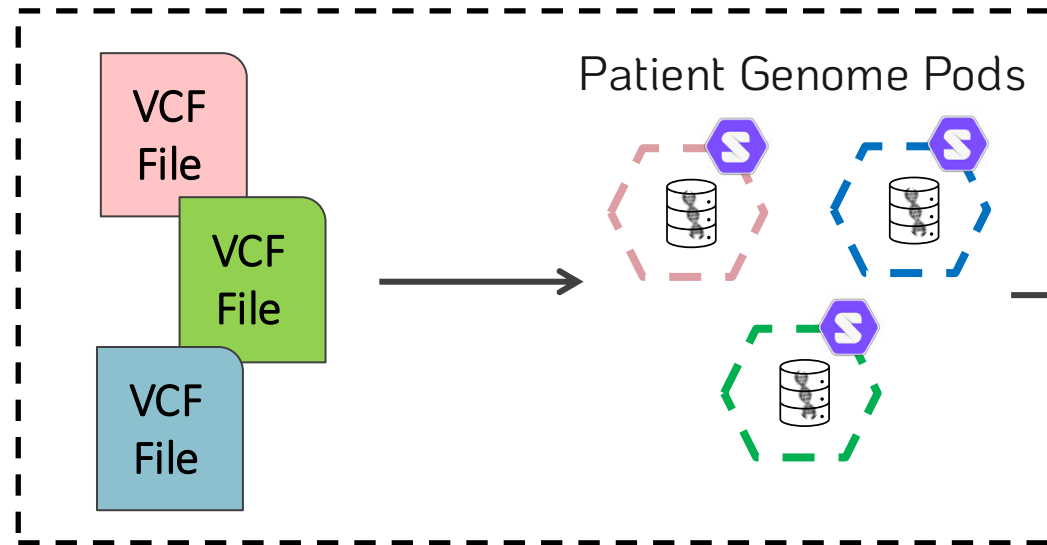


Data Ingestion

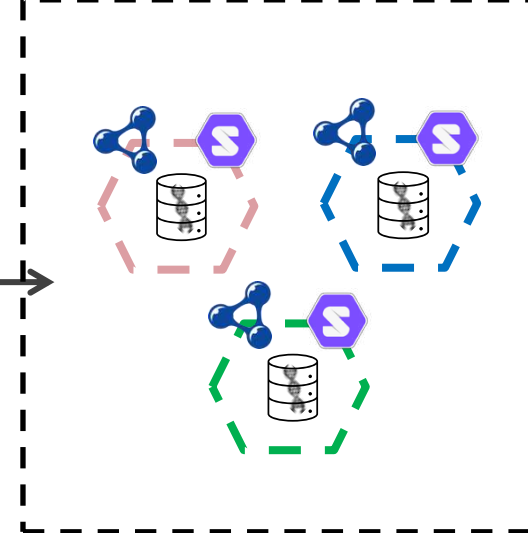


Workflow Overview

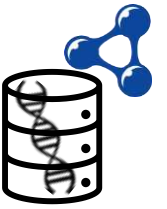
Data Storage



Representation



Genome Data in RDF

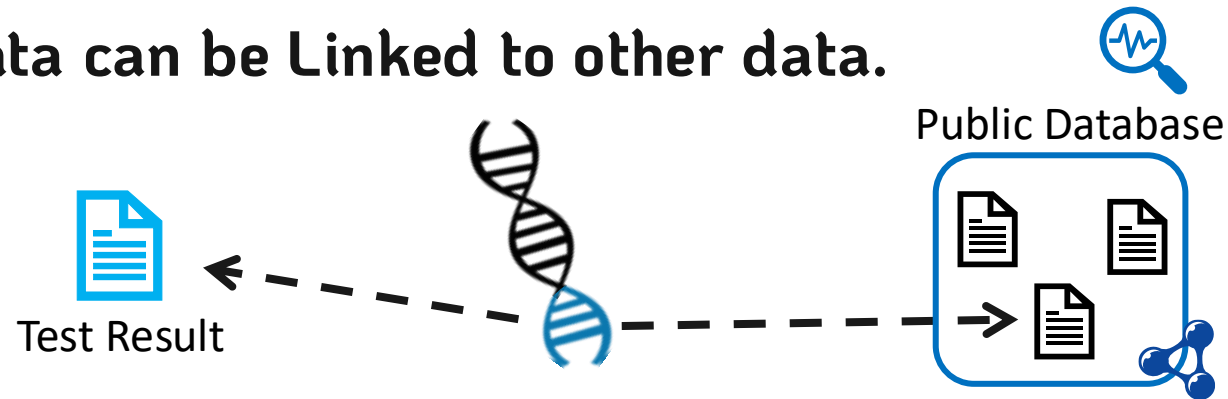


VCF genomic data can be represented as RDF.



 VCF to RDF Vocabularies exist $\longrightarrow$ SPHN Semantic Interoperability Framework⁴
+ FHIR HL7 Format⁵

VCF genomic data can be Linked to other data.



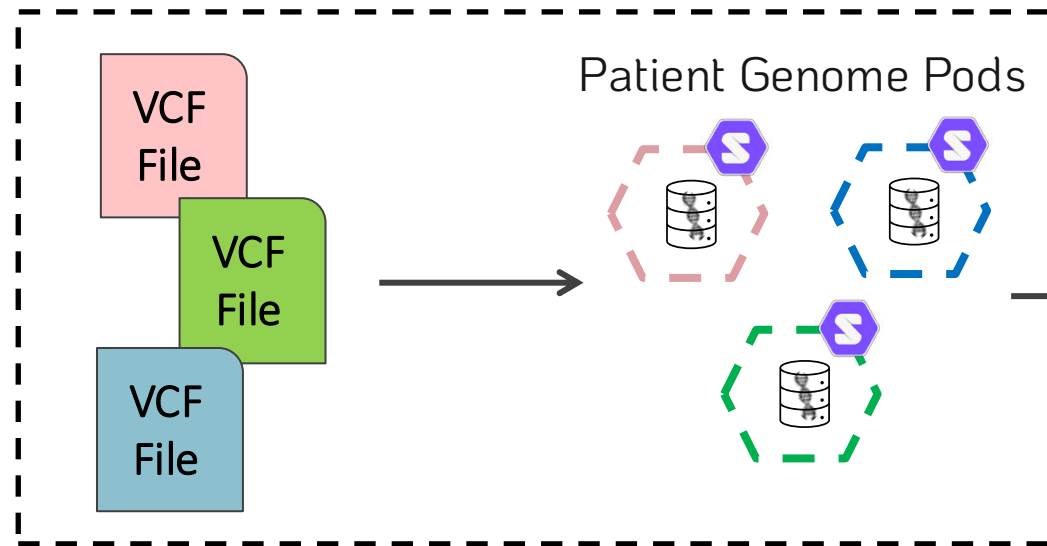
How to automate this?

What vocabularies could be used for the linkages?

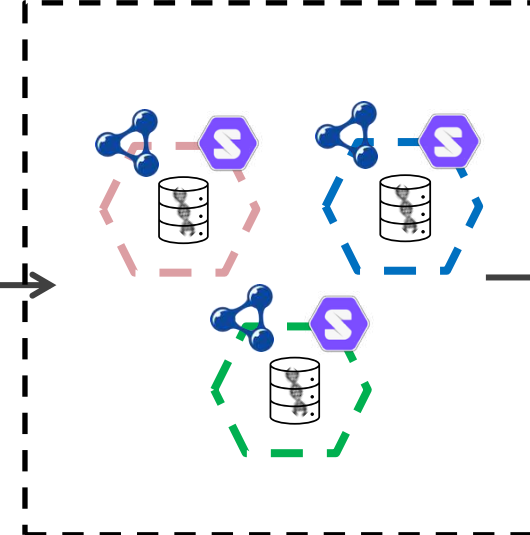


Workflow Overview

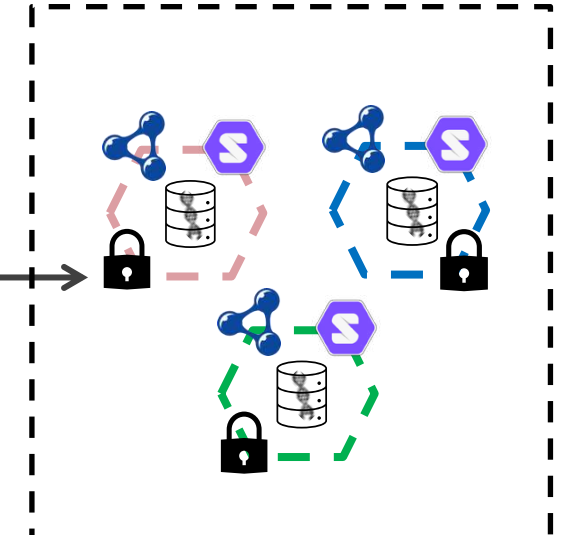
Data Storage



Representation



Privacy



Data Privacy Controls



Granular privacy controls can be implemented in Solid for genomic data.



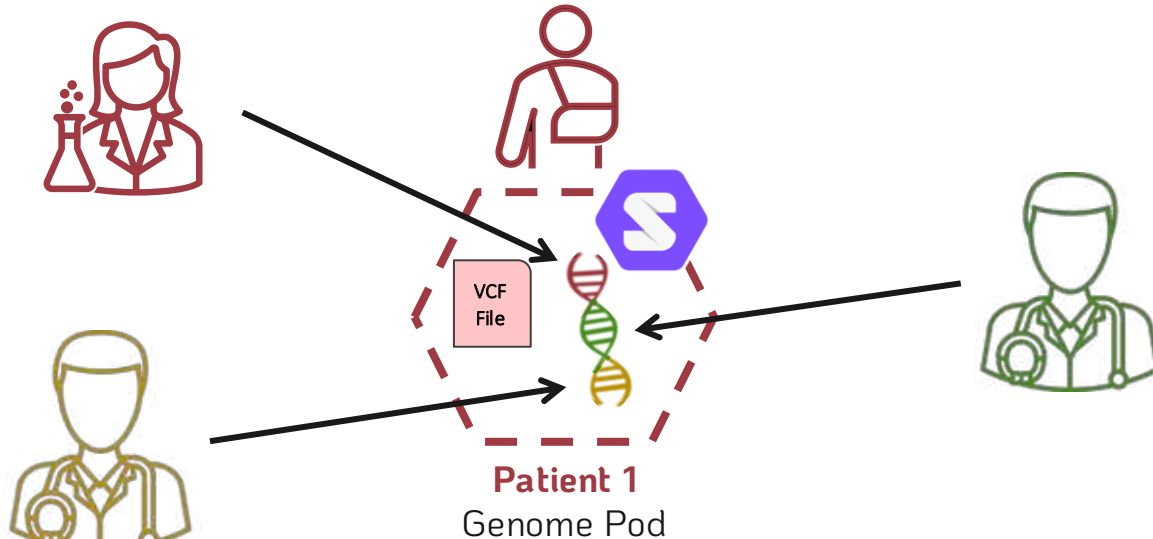
Web Access Control



Solid Protocol ACP³

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

Mutation is also a semantic Class,
above a triple but below a Chromosome

Querying Genome Pods



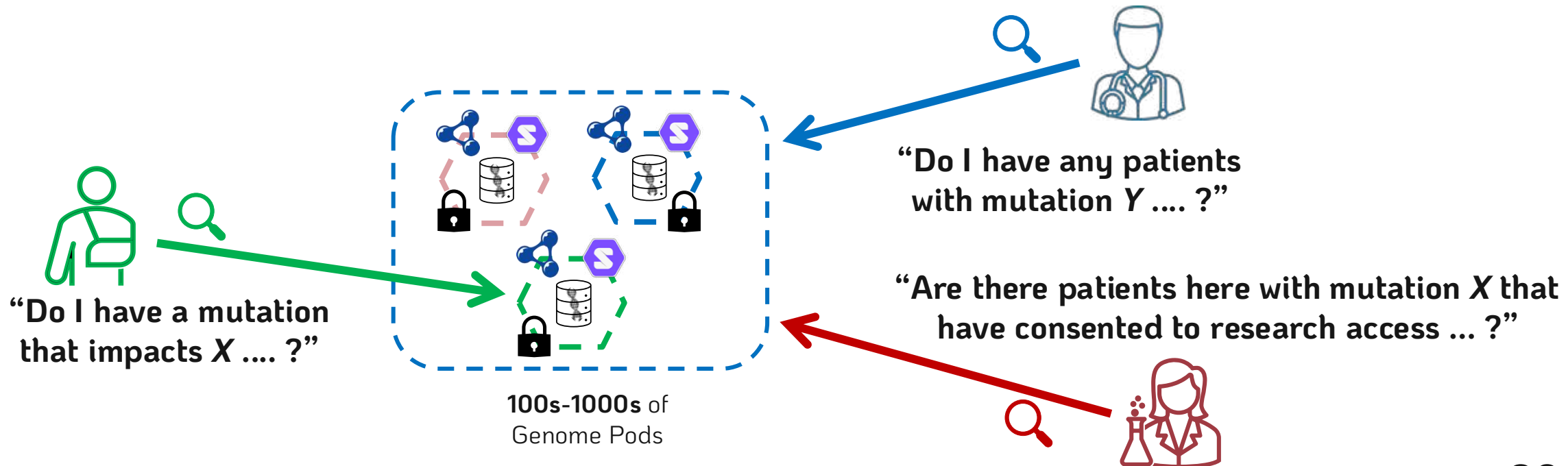
RDF genomic data pods can be queried.



Query Engine



Custom Comunica implementation/engine⁶



Querying Genome Pods (possibilities)



Single Pod Query

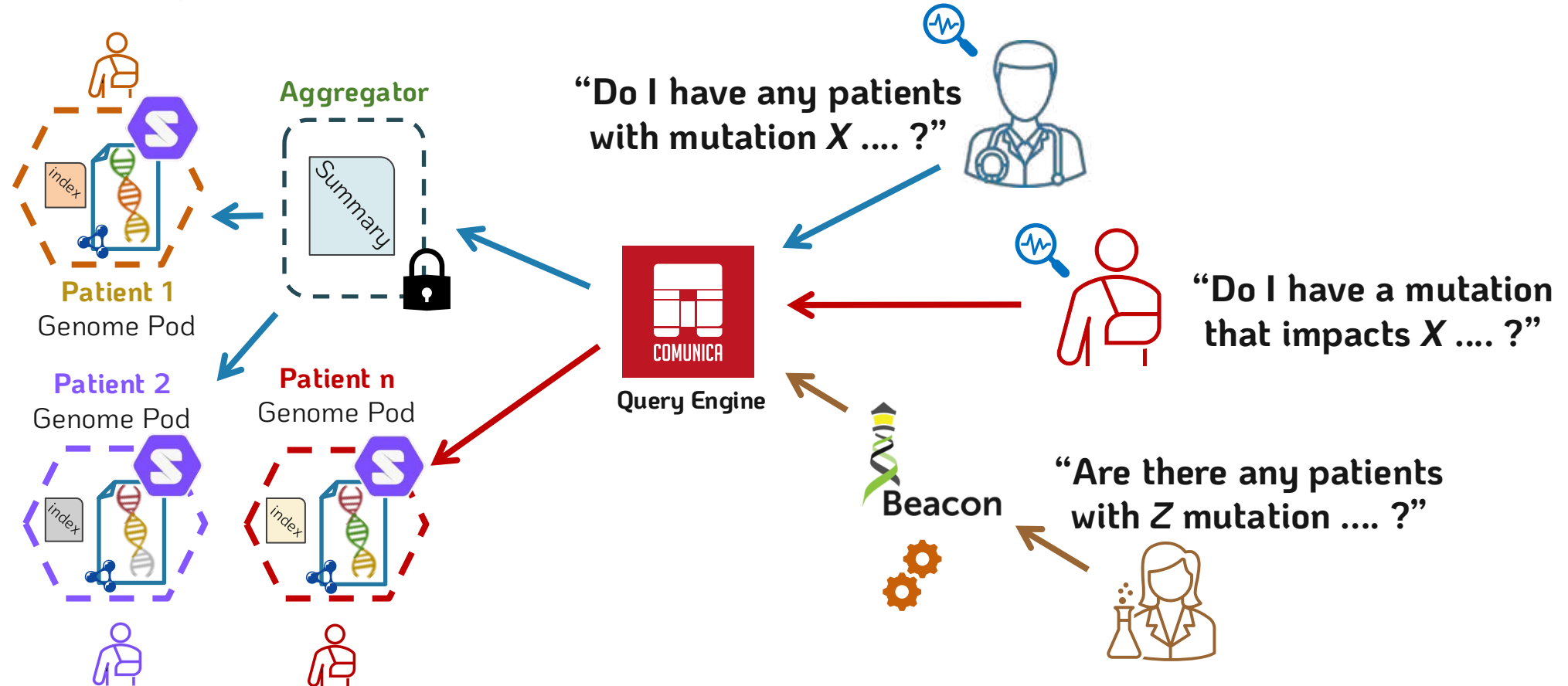
Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed



Implementation Specifics



Solid pod instances → Community Solid Server¹

Genomic Data → Publicly available *VCF* data²

RDF Conversion → Aim to use SPHN RDF vocabulary⁴

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica⁶